A PROJECT REPORT ON

Supplier Selection Scorecard

Ranking Suppliers on Cost Efficiency, Delivery Reliability and Supply Chain Resilience

Submitted for the course

FinTech Data Science and Analytics

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ABSTRACT

Companies that buy goods from many suppliers must decide again and again which supplier should get the next order. In practice, this choice is often made by looking at price alone. But a supplier that is a little cheaper and often late can end up costing more, because late deliveries lead to empty shelves, rush shipping and lost sales.

In this project we built a **Supplier Selection Scorecard**: a simple, browser-based tool that scores every supplier on three things that matter — **Cost Efficiency** (is the supplier cheap compared with others selling the same kind of product?), **Delivery Reliability** (does the supplier deliver on the promised date?) and **Resilience** (does the supplier stay steady over time?). These three scores are combined into one final score from 0 to 100, and the user can change how much each part counts.

We tested the tool on the USAID Supply Chain Shipment Pricing dataset, which has 10,324 shipment records and 73 vendors. After cleaning, 25 vendors had enough shipments (at least 20) to be ranked fairly. The results show clearly why price alone is a poor guide. For example, Aurobindo Pharma and Cipla are among the six cheapest vendors, but they fall to 22nd and 24th place out of 25 once delivery record is taken into account, because only about 86–87% of their shipments arrived on time. We also tested how stable the ranking is when the weights change, and found that the top two suppliers stay in the top three in every case.

The project shows that a simple, transparent scoring method can give procurement teams a much better picture of supplier quality than price alone, and it gives a base that can later be extended with more advanced methods such as AHP, TOPSIS and optimisation.

Keywords: supplier selection, scorecard, multi-criteria decision making, delivery reliability, supply chain resilience, FinTech, data analytics

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1. Introduction

1.1 Background

Almost every business buys things from other businesses. A hospital buys medicines, a shop buys stock, a factory buys raw material. When there are several suppliers who can deliver the same item, someone has to decide who gets the order. This is called **supplier selection**.

In many companies this decision is still made on gut feeling, or by picking the lowest price. Price is easy to see and easy to compare, so it becomes the main number people look at. But price is only one part of the real cost of buying from a supplier.

1.2 Why This Problem Matters

- **Too many suppliers to compare by hand.** Procurement teams often deal with 50 to 500 or more active suppliers. Comparing all of them manually, order by order, is slow and full of mistakes.
- **The cheapest supplier is not always the cheapest choice.** A supplier who is 5% cheaper but late much more often can cost more overall. Late deliveries cause stock-outs, extra rush shipping and lost sales.
- **The world has become less predictable.** Events like COVID-19, the Suez Canal blockage and sudden tariff changes have shown that supply chains can break. Companies that focus only on low price are often the ones hit hardest.

So there is a real need for a method that looks at price *and* delivery *and* stability together, and gives one clear answer that people can trust and understand.

1.3 An Easy Way to Think About It: Signing a Football Player

Our idea is similar to how a football club decides which player to sign. A club does not simply buy the cheapest player. It looks at many things — how much value the player brings, how much fans trust him, and whether he helps the team win — and then combines them into one overall “transfer value”. We do the same for suppliers.

Table 1: Football transfer analogy mapped to supplier selection

Football transfer model	Supplier scorecard model
Player	Supplier
Transfer fee / wages	Unit cost / total landed cost
Profitability (return for the club)	Cost Efficiency — price compared with a fair benchmark
Popularity (fan trust)	Delivery Reliability — on-time delivery, our “trust” measure
Win probability (helps the team win)	Resilience — steady performance even when things get hard
Composite “transfer value”	Composite supplier score
Which player to sign	Which supplier to buy from

1.4 Link to FinTech

Although the data comes from a supply chain, the thinking behind the project is very close to finance. Choosing a supplier is like choosing an investment: each supplier offers some “return” (savings on price) and carries some “risk” (late or unpredictable delivery). A good decision balances both, instead of chasing return alone. We explain this link in more detail in Chapter 9.

1.5 How This Report Is Organised

Chapter 2 states the problem and our goals. Chapter 3 gives a short background on how supplier selection is usually studied. Chapter 4 describes the dataset and its problems. Chapters 5 to 7 explain how our system is designed, the method behind the scores, and how we built it. Chapter 8 shows and discusses the results. Chapter 9 explains the FinTech view. Chapters 10 to 12 cover limitations, future work and the conclusion.

2. Problem Statement and Objectives

2.1 Problem Statement

Given past data on orders, shipments, prices and delivery dates, build a model that scores and ranks suppliers on **Cost Efficiency**, **Delivery Reliability** and **Supply Chain Resilience**, and combines these into a single scorecard that a procurement team can use to choose suppliers or shift orders between them.

2.2 Questions We Want to Answer

- Which suppliers are really cheap once we compare them fairly with others selling the same type of product?
- Which suppliers deliver on time most often, and how late are they on average?
- Which suppliers stay steady over time, instead of being good one month and bad the next?
- How should the three areas be weighted, and does the final ranking change a lot if the weights change?
- How different is our ranking from simply ranking suppliers by price?

2.3 Objectives

- To clean a real-world shipment dataset that contains messy values such as text inside number columns.
- To turn the words “cost efficiency”, “reliability” and “resilience” into clear, measurable numbers.
- To combine these numbers into one fair score for each supplier, with weights that the user can adjust.
- To check how stable the ranking is by testing different weight settings (sensitivity analysis).
- To compare our ranking with a simple price-only ranking and show the difference.
- To build all of this into an easy-to-use web app that anyone can run without installing anything.

2.4 Scope

The project covers data cleaning, metric design, scoring, ranking, sensitivity testing and comparison with a price-only baseline. More advanced parts such as AHP/TOPSIS weighting, an optimisation layer and back-testing on a later time period are discussed as future work (Chapter 11) and are not part of the current build.

3. Background Study

3.1 Supplier Selection as a Multi-Criteria Decision

Supplier selection has been studied for a long time. One of the earliest well-known studies, by Dickson (1966), found that buyers care about many things at once — quality, delivery, price, past performance and more — not just price. Since then, supplier selection has been treated as a **multi-criteria decision-making (MCDM)** problem, where several different measures must be combined into one choice (Ho, Xu and Dey, 2010).

3.2 Delivery Reliability and the SCOR Model

The **SCOR model** (Supply Chain Operations Reference), a widely used industry framework, treats "reliability" as one of the key features of a supply chain. A common measure is **OTIF** — On Time, In Full — which checks whether an order arrived on the promised date and in the right quantity. Our on-time rate follows the same idea, using the promised and actual delivery dates in the data.

3.3 Common Ways to Combine Criteria

- **Simple weighted average:** each criterion gets a weight and the final score is the weighted average. Easy to explain and easy to change. This is what our tool uses today.
- **AHP (Analytic Hierarchy Process):** created by Saaty (1980). The decision maker compares criteria two at a time (for example, "is cost more important than reliability?") and the method works out the weights from these answers.
- **TOPSIS:** ranks each option by how close it is to an "ideal" supplier (best on everything) and how far it is from the "worst" one (Hwang and Yoon, 1981).

3.4 Where Our Project Fits

Many research papers apply AHP or TOPSIS to small, made-up examples with a handful of suppliers. Our project takes a more hands-on approach: it works with a large, real and messy dataset, builds clear metrics from raw dates and prices, and puts everything into a working tool. We start with the simple weighted average because it is the easiest for a non-technical user to understand, and we test its stability with sensitivity analysis.

4. Dataset

4.1 Data Source

We used the **USAID Supply Chain Shipment Pricing Data** (also known as SCMS Delivery History), available on Kaggle. It contains records of shipments of HIV medicines (ARV) and HIV test kits sent to many countries. We picked this dataset because it has a real, named **Vendor** column, which is exactly what a supplier scorecard needs. A second dataset, the DataCo Smart Supply Chain data, was also considered, but it has no supplier column and would have needed a made-up “supplier” unit, so we kept it for future work.

Table 2: Dataset at a glance

Item	Value
Number of shipment records	10,324
Number of columns	33
Number of unique vendors	73
Scheduled delivery period	May 2006 to December 2015
Vendors with at least 20 shipments (ranked)	25
Freight cost values that are text, not numbers	4,126

4.2 Columns Used

Table 3: Dataset columns and how we use them

Column	Role in our model
Vendor	The supplier. Each unique vendor becomes one row on the scorecard.
Scheduled Delivery Date	The date the vendor promised to deliver.
Delivered to Client Date	The date the goods actually arrived.
Unit Price	Used to build the price index (cost).
Product Group	Makes price comparison fair (like-for-like).
Freight Cost (USD)	Shipping cost, used for the freight ratio.
Line Item Value	Total value of the order line, used for the freight ratio.

4.3 A First Look at the Data

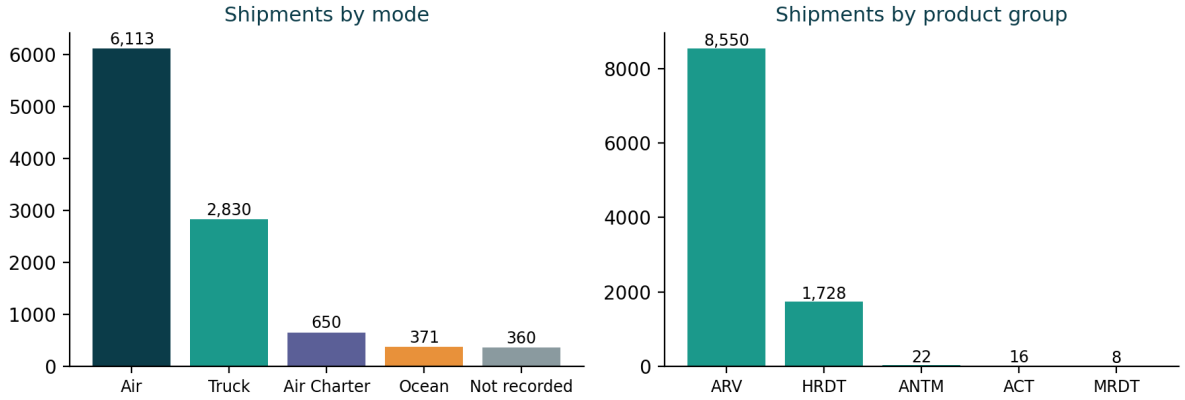


Figure 1: Shipments by mode and by product group. ARV = HIV medicines, HRDT = HIV rapid test kits

Most shipments went by air, and most were HIV medicines (ARV). Across all records, **88.5%** of shipments arrived on or before the scheduled date, and 1,186 shipments were late. The single largest "vendor" is *SCMS from RDC*, with 5,404 shipments; this is actually the programme's own regional distribution centre rather than an outside company, which we note as a limitation.

4.4 Data Quality Problems

Real data is rarely clean. The main problems we found were:

- **Text inside number columns.** The Freight Cost column has entries like "Freight Included in Commodity Cost" (1,442 rows), "Invoiced Separately" (239 rows) and references such as "See ASN-32231". The Weight column has "Weight Captured Separately" (1,507 rows). In total, 4,126 freight values are not numbers.
- **Unusual date formats.** Dates are written like "2-Jun-06", which many tools do not read correctly by default.
- **Special characters.** Some country names (for example "Côte d'Ivoire") use characters that break if the file is opened with the wrong text encoding.
- **Very small vendors.** Many vendors have only a few shipments. One late delivery would swing their score a lot, so their scores would not be fair.

How we handled each of these is explained in Section 5.4.

5. System Design

5.1 Overall Workflow

The tool is a web app that walks the user through five simple steps, from uploading a file to seeing the final ranking.

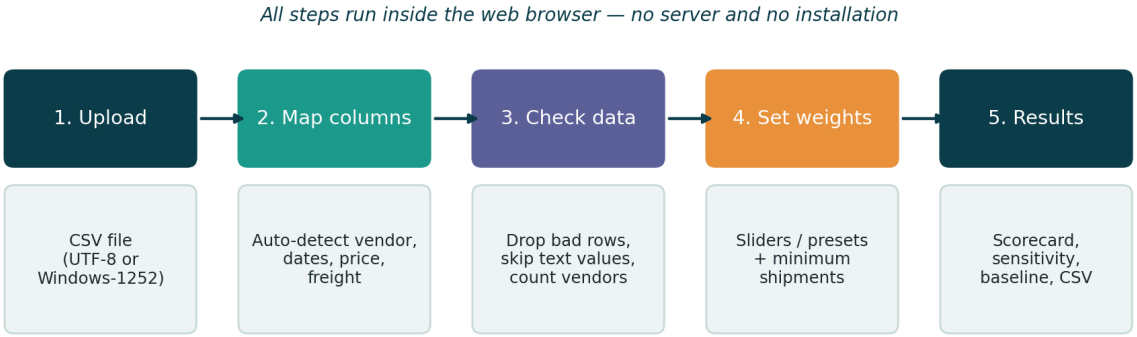


Figure 2: Five-step workflow of the Supplier Scorecard app

5.2 Architecture

The whole app runs inside the web browser. There is no server and nothing to install — the user just opens *index.html* in Chrome, Firefox or Edge. A useful side effect is privacy: the uploaded data never leaves the user's computer.

Table 4: Project files

File	What it does
index.html	Page layout and structure of the app.
style.css	Colours, fonts and visual design.
script.js	All the logic: reading the CSV, cleaning, calculating metrics, scoring, ranking and showing results.
PapaParse (library)	A trusted open-source library used to read CSV files correctly, including quoted text with commas.
sample-data.csv	The USAID dataset, included for testing.

5.3 Automatic Column Mapping

Different datasets name their columns differently. To make the app work with any shipment file, it guesses which column is which. It first looks for exact known names (for example "Vendor" or "Scheduled Delivery Date"). If none is found, it searches by keywords while skipping clearly wrong columns — for example, "Vendor INCO Term" contains the word "vendor" but is not the supplier name, so it is skipped. Each column can be used for only one role, and the user can correct any guess from a drop-down list.

5.4 Data Cleaning Steps

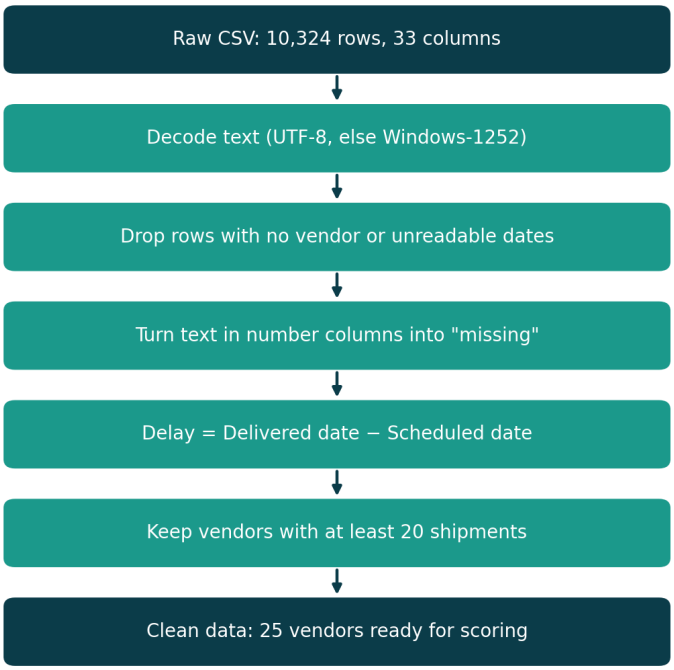


Figure 3: Data cleaning pipeline

- **Encoding:** the file is read as UTF-8 first. If that fails, it is read as Windows-1252, so special characters show correctly.
- **Dropping bad rows:** rows without a vendor name or with dates that cannot be read are removed and counted, so the user can see what was dropped.
- **Dates:** formats such as “2-Jun-06”, “2006-06-02” and “06/02/2006” are all understood.
- **Text in number columns:** values like “Freight Included in Commodity Cost” are treated as *missing*, not as zero. Treating them as zero would wrongly make a supplier look like it has free shipping.
- **Minimum shipments:** vendors with fewer than 20 shipments are left out of the ranking (the user can change this number).

6. Methodology

This chapter explains how raw shipment records are turned into a single score for each supplier. Figure 4 shows the full process at a glance.

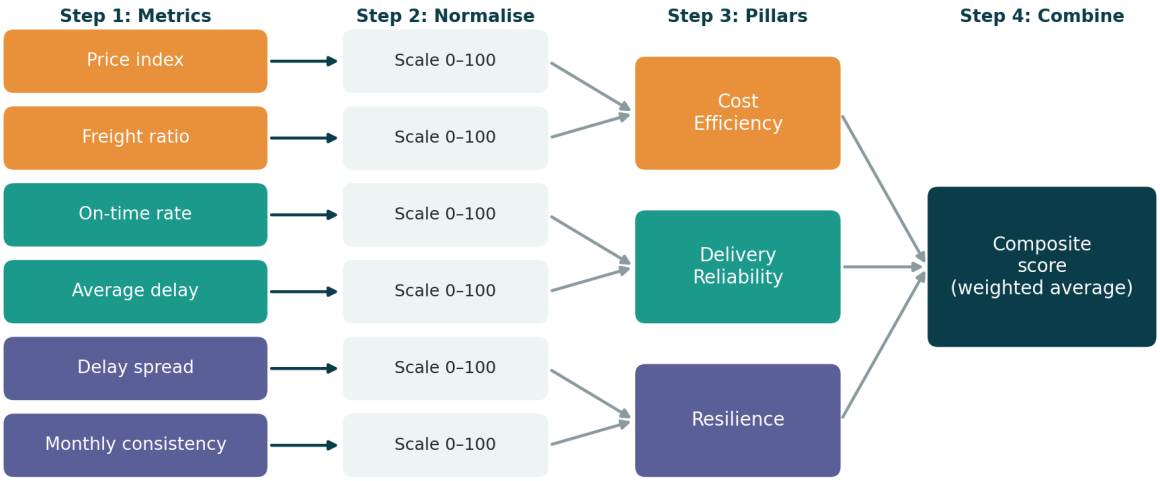


Figure 4: From six metrics to three pillars to one composite score

6.1 Step 1 — Metrics for Each Supplier

For every supplier, we first calculate the delay of each shipment:

$$\textit{Delay (days)} = \textit{Delivered Date} - \textit{Scheduled Delivery Date}$$

A negative delay means the shipment arrived early; zero means exactly on time. From this and the price data, we calculate six metrics:

Table 5: The six metrics used in the scorecard

Pillar	Metric	How it is calculated	Better when
Cost	Price index	Supplier's average unit price ÷ median unit price of the same product group	Lower
Cost	Freight ratio	Total freight cost ÷ total line item value	Lower
Reliability	On-time rate	Share of shipments with delay ≤ 0	Higher
Reliability	Average delay	Mean of all delays, in days	Lower
Resilience	Delay spread	Standard deviation of delay	Lower
Resilience	Monthly consistency	Standard deviation of the on-time rate across months (months with at least 3 shipments)	Lower

The **price index** is the key idea for fair cost comparison. A vendor selling test kits should not look "cheap" just because test kits cost less than medicines. By dividing by the median price of the *same product group*, a value of 1.0 means "typical price", below 1.0 means cheaper than typical, and above 1.0 means more expensive.

6.2 Step 2 — Putting Everything on the Same Scale

The six metrics are in different units (days, ratios, percentages), so they cannot be added directly. We use **min-max scaling** to convert each one to a 0–100 scale across all suppliers:

$$\text{Scaled value} = (x - \text{min}) \div (\text{max} - \text{min}) \times 100$$

For metrics where lower is better (like delay or price), we flip the result so that **100 always means best**:

$$\text{Scaled value} = (1 - (x - \text{min}) \div (\text{max} - \text{min})) \times 100$$

6.3 Step 3 — Pillar Scores

Each pillar score is the simple average of its scaled metrics. For example, the Reliability score is the average of the scaled on-time rate and the scaled average delay. If a metric is missing for a supplier (for example, no usable freight data), the pillar uses whatever metrics are available.

6.4 Step 4 — Composite Score and Weights

The final score is a weighted average of the three pillar scores:

$$\text{Composite} = (w_C \times \text{Cost} + w_R \times \text{Reliability} + w_S \times \text{Resilience}) \div (w_C + w_R + w_S)$$

The user sets each weight from 0 to 5 using sliders. Weights are relative, so 1/1/1 and 2/2/2 give the same result. A weight of 0 removes that pillar. If a whole pillar has no data, it is dropped and the other weights are adjusted automatically. The default is equal weights (1/1/1).

6.5 Step 5 — Sensitivity Analysis

A ranking that changes completely when the weights move a little is not safe to act on. So the tool re-ranks every supplier under four preset weight settings and shows the results side by side:

Table 6: Weight scenarios used for sensitivity analysis

Scenario	Cost : Reliability : Resilience	Meaning
Equal	1 : 1 : 1	All three areas matter the same.
Cost-first	3 : 1 : 1	Price matters three times as much.
Reliability-first	1 : 3 : 1	On-time delivery matters most.
Resilience-first	1 : 1 : 3	Steady, predictable performance matters most.

6.6 Step 6 — Comparison with a Price-Only Baseline

Finally, the tool also ranks suppliers by price index alone. This is the “naive” method many buyers use today. Putting both rankings side by side shows exactly which suppliers look good on price but fall down once delivery and stability are counted.

7. Implementation

7.1 Technology Used

The app is written in plain **HTML, CSS and JavaScript**, with the **PapaParse** library for reading CSV files. We chose this simple stack on purpose: it runs on any computer with a browser, needs no setup, and is easy for evaluators to test. For preparing the figures and checking the numbers in this report, we used **Python** with pandas and matplotlib, applying exactly the same method as the app.

7.2 Main Functions in script.js

Table 7: Key functions and what they do

Function	Purpose
autoMap()	Guesses which column is the vendor, the dates, the price and so on.
parseDate() / toNum()	Read dates in many formats; turn text into numbers or mark them missing.
buildClean()	Removes bad rows, calculates each shipment’s delay and counts data problems.
supplierMetrics()	Groups shipments by vendor, applies the minimum-shipment rule and calculates the six metrics.
scoreAll()	Scales metrics to 0–100, builds pillar scores and the weighted composite score, then sorts.
runAll()	Runs scoring under the chosen weights, the four sensitivity scenarios and the price-only baseline.
exportCsv()	Lets the user download the ranked scorecard as a CSV file.

7.3 User Interface

The interface follows the five steps shown in Figure 2, with a step bar at the top. Each screen explains itself in plain language: for example, the column-mapping screen explains what each column is used for, and the data-check screen shows how many rows were dropped and why. The results screen has three parts — the ranked scorecard with a small bar for each score, the sensitivity table, and the price-only baseline — plus a button to download the results.

7.4 Handling Edge Cases

- If a file is not a CSV, the app shows a clear message instead of crashing.
- Numbers with commas or dollar signs (for example “$1,200”) are read correctly.
- If every supplier has the same value for a metric, all get 100 for that metric, to avoid dividing by zero.
- If no supplier meets the minimum shipment count, the app asks the user to lower the limit.

8. Results and Discussion

8.1 Test Setup

We ran the scorecard on the full USAID dataset with **equal weights (1:1:1)** and a **minimum of 20 shipments** per vendor. All 10,324 rows had readable dates and a vendor name, so no rows were dropped. Out of 73 vendors, 25 met the 20-shipment rule and were ranked; the other 48 were left out because their scores would be too noisy.

8.2 Final Ranking

Table 8: Full supplier scorecard (equal weights, minimum 20 shipments). Ships = number of shipments; Reliab. = Reliability; Resil. = Resilience

#	Vendor	Ships	Score	Cost	Reliab.	Resil.	On-time	Avg delay (d)
1	Bristol-Myers Squibb	67	82.3	89.8	57.1	100.0	100.0%	0.0
2	Pharmacy Direct	326	82.3	89.7	57.1	100.0	100.0%	0.0
3	Hoffmann-La Roche	23	80.6	84.7	57.1	100.0	100.0%	0.0
4	GlaxoSmithKline Export	20	80.0	82.8	57.1	100.0	100.0%	0.0
5	Mylan Laboratories	317	79.1	90.6	56.2	90.4	99.4%	-0.6
6	Hetero Labs	277	79.0	87.0	55.3	94.5	99.3%	-0.2
7	Aspen Pharmacare	41	78.0	89.8	48.7	95.5	97.6%	0.9
8	Micro Labs	35	77.1	74.3	57.1	100.0	100.0%	0.0
9	Trinity Biotech	356	76.5	80.1	56.2	93.3	99.7%	0.1
10	AbbVie Logistics	347	76.1	86.3	53.3	88.6	98.8%	0.3
11	Abbott GmbH	21	75.4	94.7	39.8	91.6	95.2%	2.3
12	Chembio Diagnostics	109	72.8	86.0	60.6	71.7	99.1%	-4.1
13	Merck Sharp & Dohme	68	71.2	90.0	54.5	68.9	98.5%	-1.1
14	JSI R&T Institute	38	70.4	91.5	72.2	47.5	97.4%	-15.2
15	Standard Diagnostics	98	69.6	79.8	44.9	84.1	95.9%	0.3
16	Strides Arcolab	93	68.8	93.1	43.1	70.2	95.7%	1.0
17	OraSure Technologies	56	63.8	68.3	51.2	72.0	98.2%	0.5
18	S. Buys Wholesaler	715	63.3	46.1	95.5	48.2	98.5%	-28.7
19	Emcure Pharmaceuticals	41	58.2	89.3	40.8	44.6	95.1%	1.4
20	Orgenics	754	57.3	88.9	18.0	65.0	87.0%	0.8
21	Bio-Rad Laboratories	28	55.4	65.3	12.4	88.5	85.7%	2.1
22	Aurobindo Pharma	668	51.7	90.4	9.5	55.1	85.9%	4.4
23	SCMS from RDC	5404	51.3	92.1	19.5	42.4	82.8%	-8.3
24	Cipla	175	49.3	92.0	11.7	44.3	86.9%	4.8
25	Shanghai Kehua (KHB)	70	41.1	0.3	47.2	75.9	97.1%	1.1

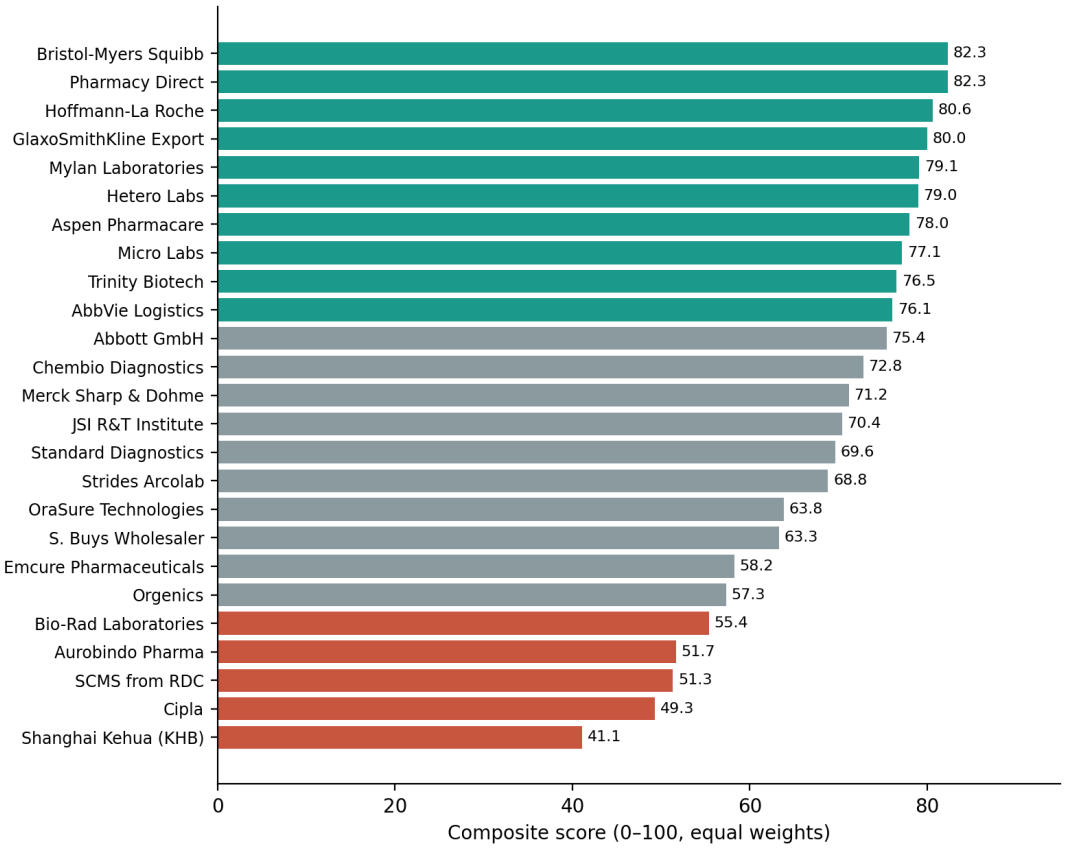


Figure 5: Composite score of all 25 ranked vendors. Teal = top 10, red = bottom 5

The top of the table is made up of vendors that delivered almost every shipment on time with very little variation, such as Bristol-Myers Squibb, Pharmacy Direct and Hoffmann-La Roche. The bottom includes vendors whose delivery record was weaker or less steady, such as Cipla and Aurobindo Pharma, and Shanghai Kehua (KHB), whose prices were far above the typical level for its product group (price index about 27).

8.3 Looking Inside the Scores

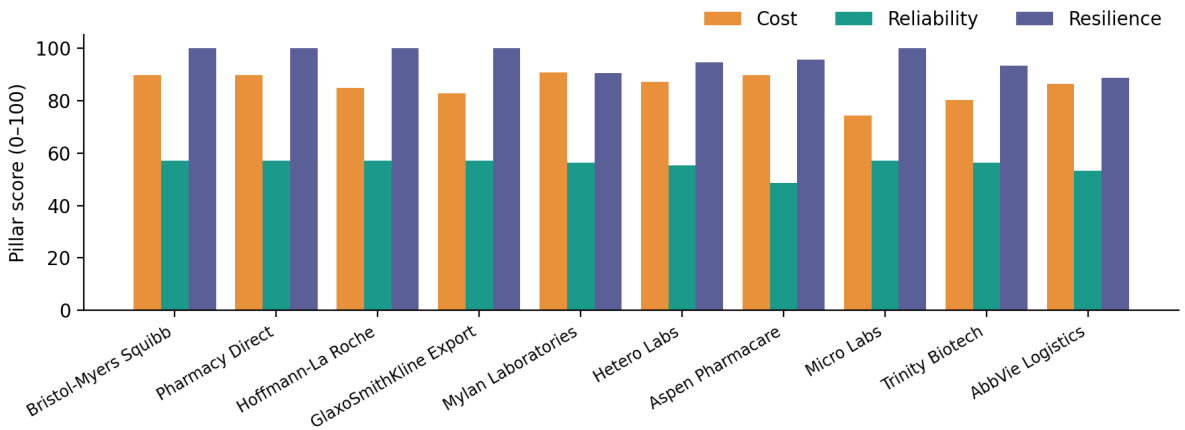


Figure 6: Pillar scores of the top 10 vendors

Figure 6 shows why looking at the three pillars separately is useful. Most top vendors score very high on Resilience because their delivery was steady. Their Reliability scores sit around 55, not near 100. This is because the average-delay metric rewards vendors who deliver *early*, and a few vendors (such as S. Buys Wholesaler, which delivered about 29 days early on average) set the top of that scale.

This points to an interesting real-world question: **is early delivery always good?** Goods that arrive a month early need to be stored and paid for sooner, which ties up money. S. Buys Wholesaler scores very high on Reliability (95.5) but low on Resilience (48.2), because its delivery dates swing widely. The scorecard makes these trade-offs visible instead of hiding them in one number.

8.4 Price Alone vs. Our Scorecard

This is the main finding of the project. Table 9 compares the six cheapest vendors (by price index) with their position on our scorecard.

Table 9: The six cheapest vendors and where they land on the scorecard

Vendor	Price index	Price rank	Scorecard rank	On-time	Avg delay (d)
Micro Labs	0.40	1	8	100.0%	0.0
Strides Arcolab	0.50	2	16	95.7%	1.0
Standard Diagnostics	0.82	3	15	95.9%	0.3
Abbott GmbH	0.90	4	11	95.2%	2.3
Aurobindo Pharma	1.02	5	22	85.9%	4.4
Cipla	1.02	6	24	86.9%	4.8

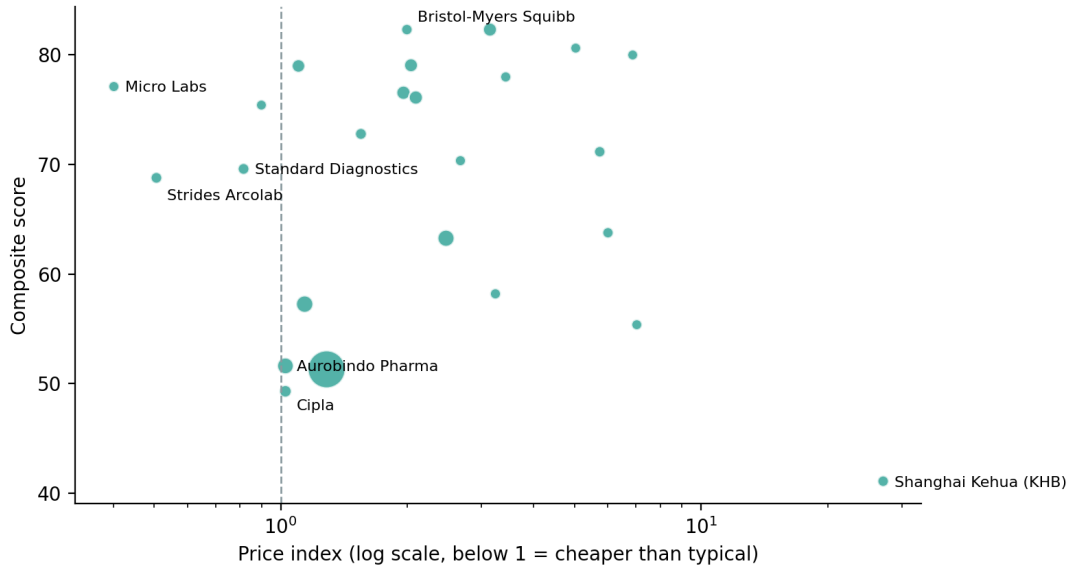


Figure 7: Price index vs. composite score. Bubble size shows number of shipments

The difference is clear:

- **Aurobindo Pharma** and **Cipla** are the 5th and 6th cheapest vendors, but they drop to **22nd and 24th** out of 25. Only about 86–87% of their shipments were on time, and they were on average 4 to 5 days late.
- **Strides Arcolab** and **Standard Diagnostics** are the 2nd and 3rd cheapest, but they fall to 16th and 15th because of weaker delivery and stability.
- **Micro Labs** is the cheapest vendor *and* delivered every shipment on time, so it stays in the top 10 (8th). Low price is not bad by itself — it just is not enough on its own.

A buyer who ranked by price alone would have sent more orders to vendors that are among the least dependable. This is exactly the kind of hidden risk the scorecard is designed to show.

8.5 Sensitivity Analysis

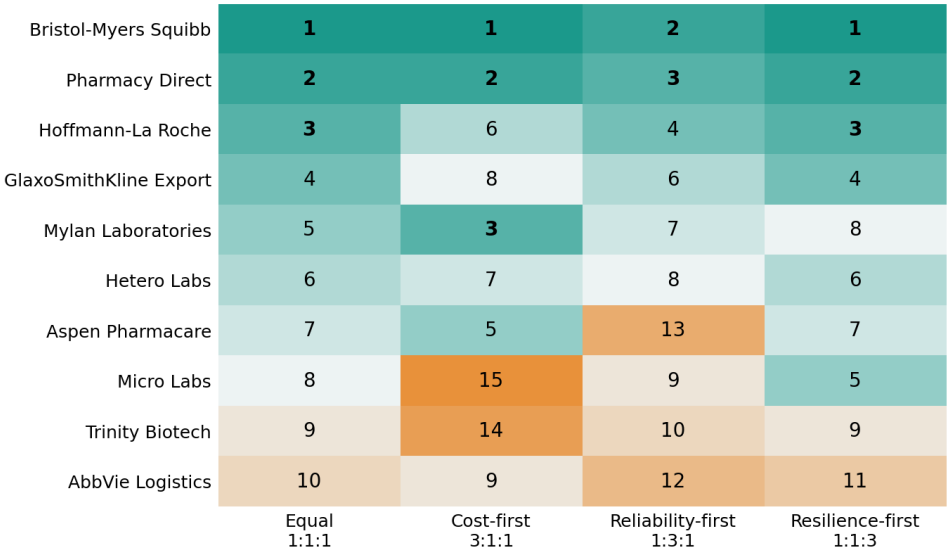


Figure 8: Rank of the top 10 vendors under four weight scenarios (1 = best)

- **The leaders are stable.** Bristol-Myers Squibb and Pharmacy Direct stay in the top 3 under every scenario. A buyer can recommend them with confidence.
- **Hidden cost shows up under Cost-first.** Micro Labs drops from 8th to 15th when cost gets triple weight. Its unit price is the lowest, but its freight cost is high (about 17% of order value), which pulls down its Cost score.
- **Some vendors depend on what you value.** Aspen Pharmacare moves from 5th (Cost-first) to 13th (Reliability-first), and Mylan Laboratories rises to 3rd under Cost-first. For these vendors, the right choice depends on the company's priorities.

8.6 Key Takeaways

1. Ranking by price alone would favour several of the least reliable vendors.
2. A few vendors are strong no matter how the weights are set — these are the safest choices.
3. Looking at the three pillars separately explains *why* a vendor ranks where it does, which makes the result easier to trust.
4. Freight cost can quietly cancel out a low unit price.

9. The FinTech View

Although this project uses supply chain data, the core idea is a financial one: **making a decision that balances return and risk**.

9.1 Suppliers as Investments

In finance, an investor does not pick a stock only because it promises the highest return. They also look at how risky it is. Markowitz's portfolio theory (1952) made this idea famous: a good portfolio balances expected return against risk. In our project:

Table 10: Mapping finance concepts to the supplier scorecard

Finance idea	Supplier scorecard idea
Expected return	Cost savings (Cost Efficiency score)
Risk / volatility	Late or unpredictable delivery (Reliability and Resilience scores)
Risk-adjusted return	Composite score
Diversification	Spreading orders across several high-scoring suppliers
Stress testing	Sensitivity analysis under different weight scenarios

9.2 The Money Behind a Late Delivery

Every late or unpredictable delivery has a cost that does not show up in the purchase price. The company may have to keep extra safety stock (money locked in inventory), pay for rush shipping, or lose sales when shelves are empty. A supplier with a slightly higher price but steady delivery can therefore be the *cheaper* choice in total. Our scorecard gives finance and procurement teams a common, data-based way to talk about this trade-off.

9.3 Hidden Costs

The Micro Labs example in Section 8.5 is a good financial lesson. Its unit price was the lowest in the dataset, but freight added about 17% on top of the order value. Looking at unit price alone would have missed this. Measuring the **total landed cost**, not just the sticker price, is a basic idea in both finance and procurement.

10. Limitations

- **Simple resilience measures.** We measure resilience using delay spread and month-to-month consistency. We do not yet test performance during specific shocks (such as peak seasons or disruptions) or across different shipment modes.
- **Manual weights.** The weights are chosen by the user. Methods like AHP or TOPSIS could give a more structured way to set them.
- **No optimisation layer.** The tool ranks suppliers but does not yet suggest how much volume to give each one.
- **No back-testing.** We have not yet checked whether top-ranked suppliers actually performed better in a later time period.
- **Dataset issues.** “SCMS from RDC” is an internal distribution centre, not an outside supplier, yet it appears as a vendor. Also, 4,126 freight values are text, so the freight ratio for some vendors is based on partial data.
- **Early delivery is always rewarded.** The average-delay metric treats very early delivery as better, although in real life it can raise storage costs.
- **One industry only.** All results come from health commodity shipments; results may differ in retail or manufacturing.

11. Future Scope

- **AHP and TOPSIS weighting:** add these well-known methods as options and compare their rankings with the simple weighted average.
- **Optimisation layer:** use linear programming (for example with PuLP or SciPy) to decide how to split an order across the top suppliers at the lowest cost while keeping a minimum reliability level.
- **Inventory link:** connect the reliability score to safety stock, so that less reliable suppliers automatically show a higher inventory holding cost.
- **Back-testing:** build the scorecard on older data and check if the top suppliers really did better in later years.
- **Richer resilience measures:** compare performance in peak vs. normal months, across air, truck and ocean shipments, and measure how quickly suppliers recover after a disruption.
- **A second dataset:** test the method on the DataCo retail dataset to show it works in other industries.
- **Better early-delivery handling:** cap the benefit of early delivery or treat very early delivery as a small penalty.

12. Conclusion

This project set out to answer a simple but important question: *which supplier should we buy from?* We showed that price alone is not a good answer. By measuring suppliers on cost, delivery reliability and resilience, and combining these into one clear score, we built a scorecard that gives a fuller and fairer picture.

Using 10,324 real shipment records from USAID, the tool ranked 25 vendors and found that some of the cheapest vendors — such as Aurobindo Pharma and Cipla — were among the least dependable. It also showed that a few vendors, like Bristol-Myers Squibb and Pharmacy Direct, stay at the top no matter how the weights are set, which makes them safe recommendations.

The tool itself is simple to use: it runs in any browser, cleans messy data on its own, explains each step in plain language and lets the user change priorities and see the effect straight away. From a FinTech point of view, it turns supplier choice into a risk-and-return decision, much like choosing investments.

There is still room to grow — adding AHP/TOPSIS, an optimisation layer and back-testing would make the model stronger. But even in its current form, the scorecard shows how data can help procurement teams make better, more balanced decisions.

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[8] DataCo Smart Supply Chain for Big Data Analysis. Kaggle dataset: shashwatwork/dataco-smart-supply-chain-for-big-data-analysis.

[9] PapaParse — fast and powerful CSV parser for JavaScript. https://www.papaparse.com

[10] Project source code: https://github.com/AlokKulkarni-11/DS-A_Project.git

Appendix A: How to Run the Project

1. Download or clone the repository from https://github.com/AlokKulkarni-11/DS-A_Project.git.
2. Open the file **index.html** in any modern browser (Chrome, Firefox or Edge). No installation or server is needed.
3. Click **Upload** and choose a shipment CSV file, for example *sample-data.csv* from the repository.
4. Check the automatic column mapping. Vendor, promised date and delivered date are required; price, product group, freight and order value are optional.
5. Review the data-check summary to see how many rows were used or dropped.
6. Set the weights with the sliders or a preset, and choose the minimum number of shipments (default 20).
7. Click **Calculate scorecard** to see the ranking, the sensitivity table and the price-only baseline.
8. Click **Download CSV** to save the results.

Output CSV Format

Table 11: Columns in the downloaded results file

Column	Meaning
rank	Position on the scorecard (1 = best)
supplier	Vendor name
composite	Final weighted score (0–100)
cost / reliability / resilience	Pillar scores (0–100)
shipments	Number of shipments used
on_time_rate	Share of on-time shipments (0–1)
avg_delay_days	Average delay in days (negative = early)